

# Basics of Computing -- Guided Notes

Unit tp-1.1 | CompTIA Network+ N10-009 Obj. FC0-U71 1.1 | N10-008 FC0-U71 1.1

Name: \_\_\_\_\_ Date: \_\_\_\_\_ Period: \_\_\_\_\_

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*Complete each question using the lesson reading. Write in the blank or answer in the space provided.*

## Question 1

A \_\_\_\_\_ is any physical component of a computer that you can touch, such as a keyboard, CPU, or hard drive. A \_\_\_\_\_ is a set of instructions that tells hardware what to do, such as an operating system or application.

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\_\_\_\_\_

\_\_\_\_\_

## Question 2

Every computer performs four fundamental functions: \_\_\_\_\_ (keyboard, mouse), \_\_\_\_\_ (CPU), \_\_\_\_\_ (RAM, hard drive), and \_\_\_\_\_ (monitor, speaker). The CPU coordinates all four of these functions.

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## Question 3

Analog signals are \_\_\_\_\_ (continuous/discrete) -- they can take any value within a range. Digital signals are \_\_\_\_\_ (continuous/discrete) -- they use only defined states such as 0 and 1. Computers process \_\_\_\_\_ data because transistors have two stable states: on and off.

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\_\_\_\_\_

\_\_\_\_\_

## Question 4

Binary is the language computers use internally because transistors have only two states: \_\_\_\_\_ (represented as 1) and \_\_\_\_\_ (represented as 0). Each binary digit is called a \_\_\_\_\_, meaning Binary digit.

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## Question 5

The CPU (Central Processing Unit) is responsible for \_\_\_\_\_ instructions. It fetches an instruction from memory, \_\_\_\_\_ it to determine what to do, and then \_\_\_\_\_ the operation. Modern CPUs perform billions of these cycles per second.

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## Question 6

The operating system (OS) is the software that manages \_\_\_\_\_ resources (CPU, RAM, storage) and provides a platform for \_\_\_\_\_ to run. Without an OS, application software cannot communicate with the hardware.

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#### Question 7

In a client-server model, the \_\_\_\_\_ device makes requests and the \_\_\_\_\_ device responds with data or services. A web browser on your laptop is an example of a \_\_\_\_\_, while the machine hosting the website is the \_\_\_\_\_.

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#### Question 8 -- Real World Application

REAL WORLD: A student asks why their smart thermostat is considered a computer even though it does not have a screen or keyboard. Using the four functions of computing, explain how the thermostat qualifies as a computer and give one example of each function in that device.

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*Download the answer key from the class server:*

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wget http://[SERVER]/resources/network-engineer/tp-1-1-basics-of-computing/tp-1-1-basics-of-computing-answer-key.pdf
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